Mini Inverter with Logic Gates

Activity Report - Digital Electronics (UEC3001)

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1. Abstract

This report describes the design, working and learning outcomes of a **mini inverter prototype** built as a Digital Electronics activity. An inverter converts DC (direct current) from a battery into AC (alternating current). In our prototype, **logic gates** generate and control the digital switching signal, **MOSFETs** switch the battery current, and a **12-0-12 centre-tapped transformer** produces the alternating output. The project shows how digital logic - NOT, NAND, AND, OR gates and flip-flops - is used in a real power-electronics application, and connects classroom theory to a working device.

2. Objectives

- To understand how a battery's DC supply can be converted into AC using electronic switching.
- To apply basic logic gates (NOT, AND, OR, NAND/NOR) in a practical circuit.
- To understand how an astable oscillator built from gates and an RC network produces a square-wave clock (about 50 Hz).
- To study complementary signals (Q and Q′) and how they drive two MOSFETs alternately in a push-pull stage.
- To learn the role of a transformer, cooling, protection and safe design practices.

- To develop teamwork, circuit-reading and documentation skills.

3. Importance of the Activity

The activity links Digital Electronics theory (gates, truth tables, flip-flops) to a working device, and shows how DC-to-AC conversion is used in backup power, UPS and solar systems. It also builds hands-on skills in PCB reading, ratings and safety, and shows CSE (DS + AI) students the hardware beneath embedded and IoT systems.

3. Components and Their Functions

Component	Function
Battery	Provides stored DC energy to run the inverter.
Transformer (12-0-12)	Transfers energy between windings and steps voltage up or down.
PCB / logic gates	Create the clock signal and control the switching that produces AC from DC.
MOSFETs and drivers	Act as fast electronic switches that connect the battery to each half of the transformer.
Output socket	Allows a suitable appliance to be connected.
Switch and display	Provide user control and a visual indication (exact function to be verified).
Cooling fan	Moves air to help prevent overheating.

4. Working Principle

The energy flows through five stages:

Battery → Logic gates → Switching (MOSFET/driver) → Transformer → AC output

1. The battery supplies a DC voltage.
2. A gate-based **oscillator** produces a square wave of roughly 50 Hz.
3. A **NOT gate** creates the opposite (complementary) signal, so there are two signals, Q and Q′.

4. Q drives **MOSFET 1** and Q′ drives **MOSFET 2**. When one is ON, the other is OFF.
5. When Q is HIGH, current flows through one half of the 12-0-12 winding. When Q′ is HIGH, current flows through the other half in the opposite direction.
6. Because the current direction keeps reversing, an alternating magnetic field is set up in the transformer, and AC appears at the output winding.

This arrangement is the typical **push-pull** topology. The block diagram is simplified, and the exact circuit sequence depends on the PCB design.

5. Basic Logic Gates and Truth Tables

NOT gate

A	Y
0	1
1	0

AND, OR and NAND gates

A	B	AND	OR	NAND
0	0	0	0	1
0	1	0	1	1
1	0	0	1	1
1	1	1	1	0

0 = LOW (0 V) and 1 = HIGH (supply voltage).

6. Logic Gates Used in the Project

Gate	Use in the mini inverter	Example
NOT	Inverts the clock to drive the second MOSFET	Q → Q′

Gate	Use in the mini inverter	Example
NAND / NOR	Builds the oscillator (astable multivibrator) with R and C	~50 Hz square wave
AND	Enable logic: run only when all conditions are OK	Switch ON AND battery OK
OR	Combines fault signals into one alarm or cut-off	Low battery OR overload
Flip-flop / counter	Divides a fast clock down to the output frequency	Clock ÷ 2

Note: the exact IC number and gates actually present on our PCB should be confirmed and entered here before final submission.

7. Applications

The prototype can supply small backup loads, such as a suitable LED light, a router or another low-power device, **only if the inverter's rating supports it**. More broadly, the same principle is used in:

- Home and office UPS units
- Solar power systems
- Vehicle inverters
- Emergency lighting
- Portable power supplies

8. Limitations

- The output is a **square wave** (or modified wave), which is not as clean as a pure sine wave and may not suit sensitive appliances.
- Output power is limited by the transformer, MOSFETs and battery.
- Efficiency is reduced by heat losses, which is why a cooling fan is used.
- The photos do not show the complete ratings or protection circuit, so the exact output voltage and power must be verified.

9. Safety Precautions

- **Mains-voltage danger:** the output may be hazardous even though the battery is low voltage. Never touch exposed wires or terminals.
- **Do not overload:** connect only loads within the verified power rating.
- **Battery care:** use the correct battery voltage and chemistry, prevent short circuits and use suitable protection.
- **Enclosure and ventilation:** keep wiring insulated, secure the enclosure and do not block the cooling fan.
- Do not assume that the output is 230 V or connect household appliances until a qualified person has verified and safely tested the design.

10. Conclusion

The mini inverter prototype demonstrates the basic idea of converting battery DC power into AC power using logic-gate-based electronic switching and a transformer. A gate-based oscillator creates a square-wave clock, a NOT gate provides complementary signals, MOSFETs switch alternately, and the 12-0-12 transformer produces the AC output. The activity linked Digital Electronics concepts such as truth tables, gate behaviour and flip-flops with a practical system, and it showed why careful attention to ratings, protection and safety is essential in real electronic design.

Input: battery (DC) → **Process:** logic gates + PCB + transformer → **Output:** AC at the socket.

11. References

1. M. Morris Mano and Michael D. Ciletti, *Digital Design*, Pearson.
2. R. P. Jain, *Modern Digital Electronics*, McGraw Hill.
3. Thomas L. Floyd, *Digital Fundamentals*, Pearson.
4. Course notes and lecture material for Digital Electronics (UEC3001).